Flood Risk Assessment for Residential Areas in Miami

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The project focuses on assessing flood risk for residential areas in Miami using geospatial techniques and multi-source datasets. By integrating river networks, elevation data, and school locations, the study first evaluates the existing flood vulnerability scenario. Spatial analysis methods such as buffering, DEM reclassification, and overlay analysis are applied to identify flood-prone zones, particularly in proximity to critical infrastructure like schools. Further, zonal statistics are used to quantify the level of risk in these areas. The project delivers data-driven insights and decision-support outputs that can help urban planners and local authorities enhance flood resilience and improve infrastructure planning in vulnerable neighborhoods.

Introduction

- 🌊 Urban flooding is a major challenge in coastal smart cities especially low -lying coastal cities like Miami
- 🛰️ Uses GIS, DEM, and hydrological datasets
- 🏫 Focus on schools near flood-prone zones
- 📍 Techniques: Buffering, Reclassification, Overlay Analysis
- 📊 Output supports planning & resilience strategies
- 🌃 Smart city approach for sustainable urban management

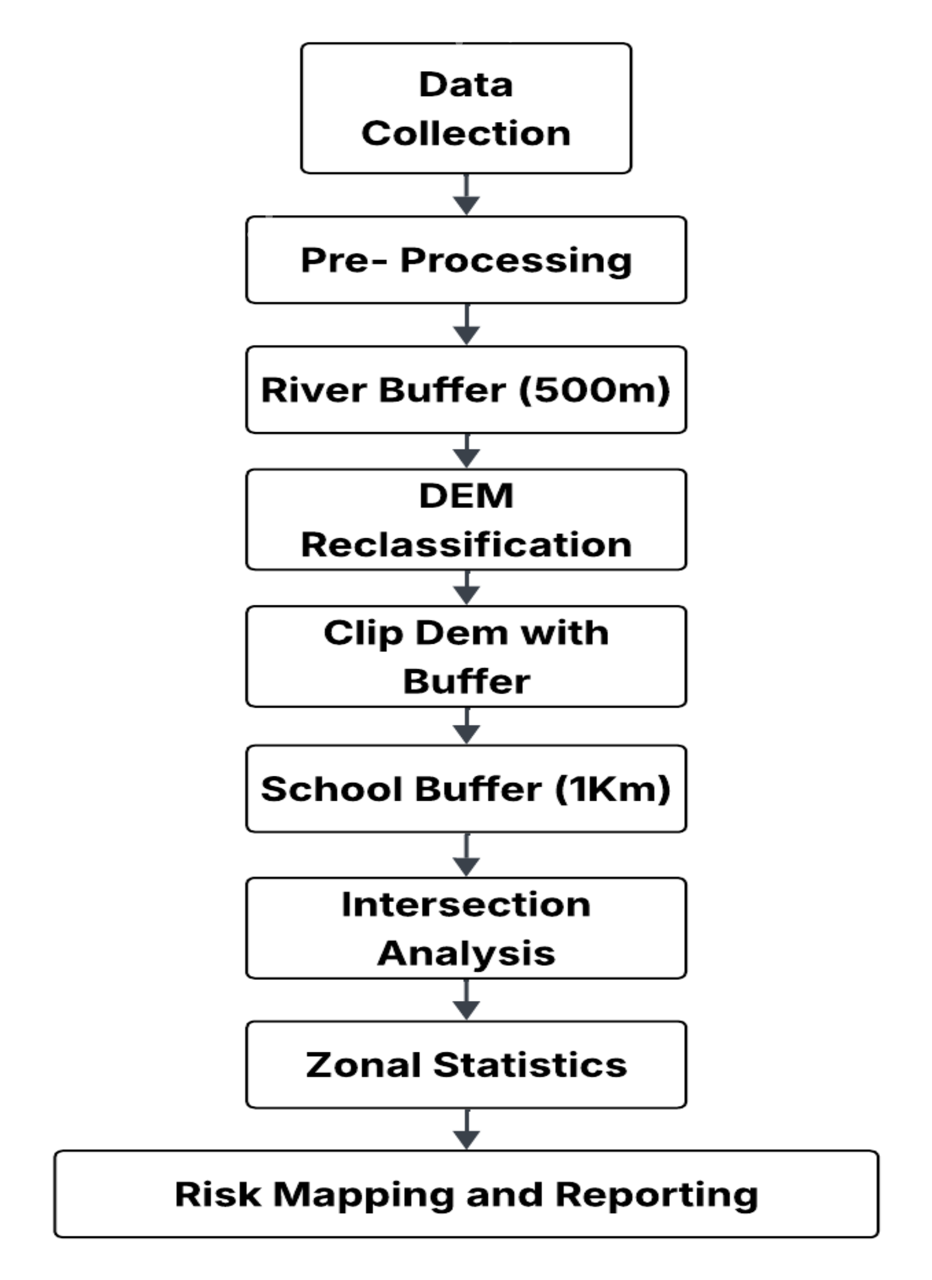
Problem Statement, Aim and Objective, Dataset

- Problem statement - Increasing urbanization and low-lying terrain in Miami have heightened flood risks near critical infrastructure like schools, necessitating accurate geospatial identification of vulnerable residential zones for effective planning and mitigation.
- Aims and objective - Identify flood prone zones near schools by buffering rivers and clipping with elevation.
- Tools used: Python, QGIS

Sno.	Dataset	Source
1	National Hydrography Dataset (NHD)	hydro.nationalmap.gov/arcgis/rest/services/nhd/MapServer.
2	DEM	USGS
3	Schools Shapefile	From MiamiDade County GIS at gis.mdc.gov/emaps/

Methodology

- Integrates hydrological, elevation, and infrastructure datasets using GIS to assess flood risk
- Buffers river networks to identify potential flood influence zones
- Uses DEM data and reclassification to detect low-lying, high-risk areas
- Overlays multiple layers to create a comprehensive flood risk map
- Intersects flood-prone zones with buffered school locations to identify vulnerable infrastructure
- Applies zonal statistics to quantify flood risk levels
- Uses spatial visualization techniques to map and interpret results
- Supports data-driven decision-making for urban planning and risk mitigation



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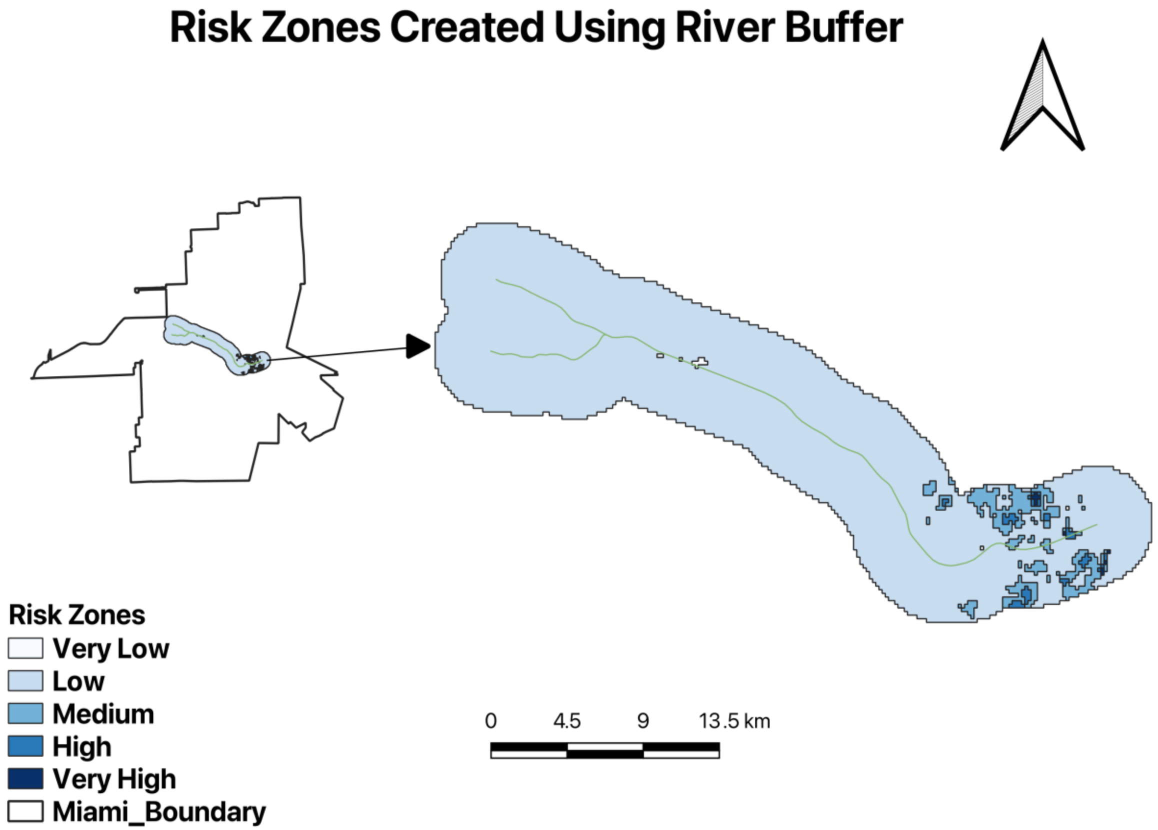
Flood Risk Zones Based on River Proximity

- **Flood risk zones are created using river buffer analysis, where areas closer to the river have higher risk.**

- **The shaded regions indicate increasing flood vulnerability, ranging from very low to very high based on distance from the river.**

- **The zoomed view highlights that high-risk zones are mainly concentrated along the river corridor within the study area.**



High-risk zones concentrated near river corridors

Flood Risk Zones Near School (Overlay Analysis)

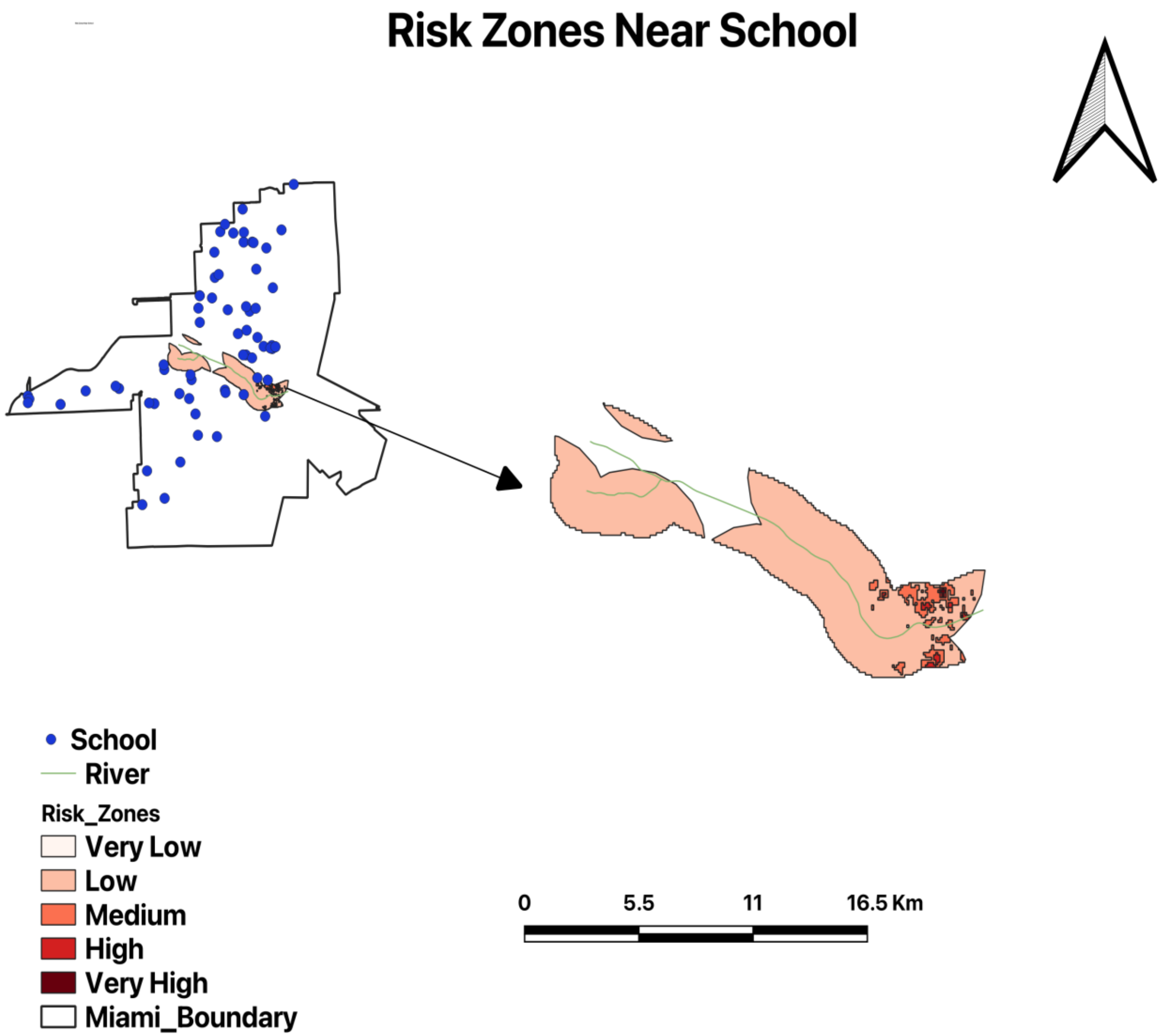
- The map illustrates Risk influenced by proximity + elevation, flood risk zones near schools in Miami, highlighting areas vulnerable due to proximity to rivers and low elevation created using intersection between river buffer and school buffer.

- Schools within buffer zones are more vulnerable

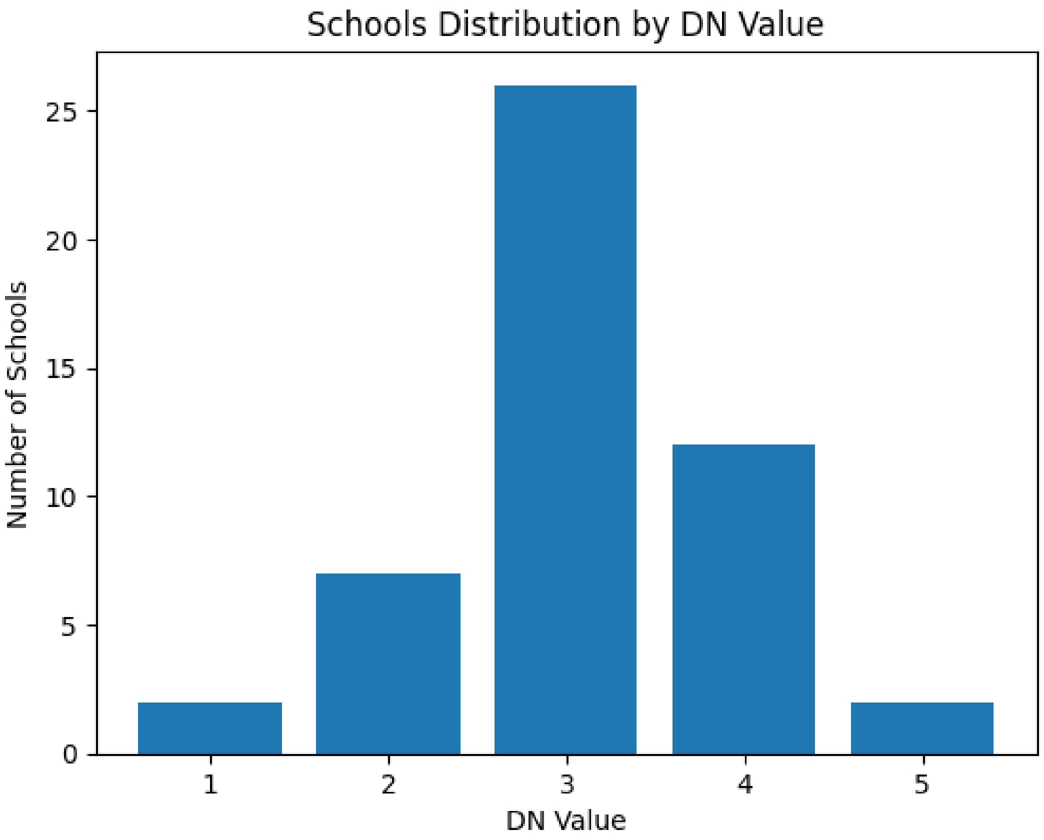
- Schools are represented as blue points, while the river network and buffered zones indicate potential flood influence areas.

- Risk zones are classified into five categories, with darker red shades representing higher flood risk along river corridors.

- The analysis identifies critical school locations within high-risk zones, supporting targeted mitigation and informed urban planning decisions.



Distribution of Schools Across Flood Risk Zones



DN	No. Of schools
1	2
2	7
3	26
4	12
5	1

Generated through python script
- https://github.com/madhubalapriya2-debug/portfolio-website

- **The table represents the distribution of schools across five flood risk zones (DN values 1 to 5).**
- **The highest number of schools (26) falls in DN = 3, indicating a moderate flood risk zone.**
- **DN = 4 has 12 schools, representing areas with higher flood risk.**
- **Very few schools are located in the lowest risk zone (DN = 1), with only 2 schools.**
- **Similarly, only 2 schools fall in the highest risk zone (DN = 5).**

Conclusion

Key Findings

- GIS-based techniques effectively identify flood-prone zones near critical infrastructure.
- Integration of river proximity, elevation data, and school locations successfully delineates high-risk areas .
- Several schools and nearby residential zones are located within vulnerable flood zones.
- High-risk zones requiring immediate attention have been clearly mapped.

Why It Matters

- Highlights potential risks to human life, especially students and residents.
- Supports data-driven decision-making for urban planning and disaster management.
- Helps prioritize areas that need urgent flood mitigation measures.
- Contributes to building resilient and sustainable urban environments.

Recommendations

- Implement targeted flood mitigation strategies in identified high-risk zones.
- Prioritize protection of critical infrastructure such as schools.
- Incorporate GIS-based flood risk analysis into urban planning policies.
- Expand and apply this scalable methodology to other flood-prone cities.
- Improve disaster preparedness and early warning systems in vulnerable areas.